Unit no. ° 7

ONE-VARIABLE DESCRIPTIVE STATISTICS

7-18 December 2026

I Counts, frequencies and mean

Definitions

A series describes a characteristic of a population. If the value x_i appears n_i times, n_i is its frequency. The non-zero total frequency is $N = \sum_{i=1}^{p} n_i$; its relative frequency is $f_i = n_i/N$. The mean is

$$\overline{x} = \frac{\sum_{i=1}^{p} n_i x_i}{N} = \sum_{i=1}^{p} f_i x_i.$$

The range is $x_{\max} - x_{\min}$.

Example 7.1: Shoe sizes · booklet data

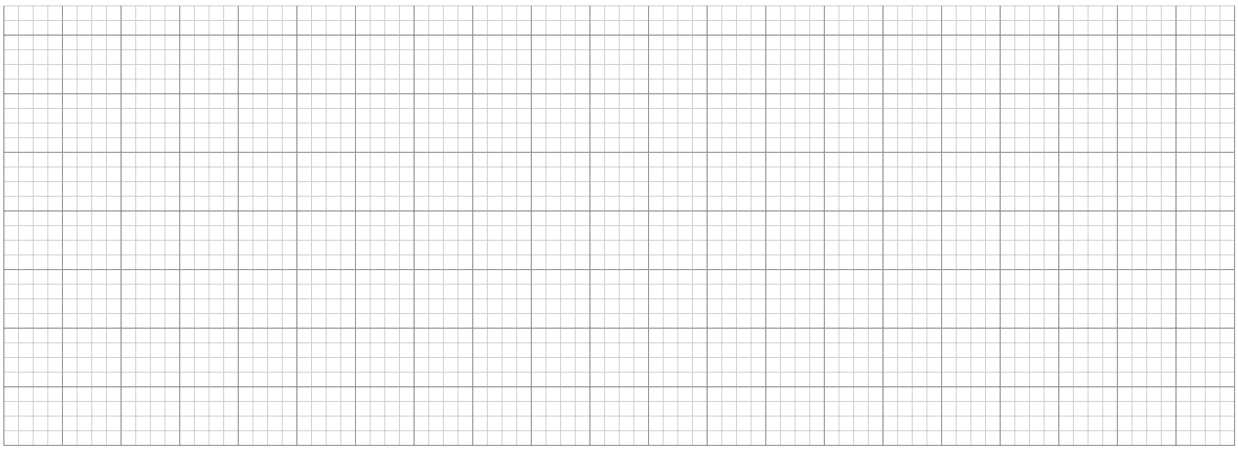
We study the shoe sizes of 30 pupils:

36	42	40	41	41	38	39	43	42	42	42	40	36	40	39
41	40	38	44	42	40	42	43	38	43	36	41	41	44	43

Complete the counts, calculate the frequencies, the mean and the range.

Shoe size	36	37	38	39	40	41	42	43	44
Count	.....	.....	.....	.....	.....	.....	.....	.....	.....

Solution:



II Median and cumulative frequencies

Definition and convention

A median is a number such that at least half the values are less than or equal to it, and at least half greater than or equal to it. In the ordered series of N values, if N is odd, take the index $(N+1)/2$; if N is even, take the mean of the values with indices $N/2$ and $N/2+1$.

Watch out. With repeated values, more than 50% of the values may be less than or equal to the median.

Example 7.2

Complete the cumulative frequencies of the shoe sizes and determine the median.

Shoe size	36	37	38	39	40	41	42	43	44
Count	3	0	3	2	5	5	6	4	2
ICC	.....	.....	.....	.....	.....	.....	.....	.....	.....

Solution:

Example 7.3

Calculate the medians of 1; 2; 2; 8; 9 and of 1; 2; 8; 9.

Solution:

III Quartiles and box plot

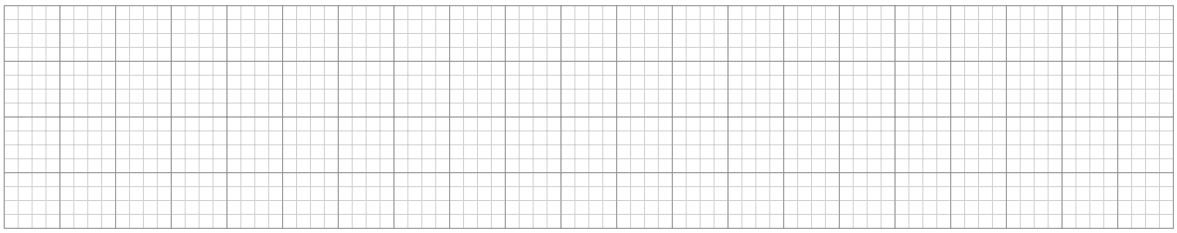
Definitions

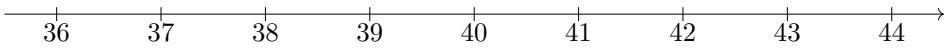
The first quartile Q_1 is the value with index $\lceil N/4 \rceil$: the smallest value in the series such that at least 25% of the values are less than or equal to it. Q_3 is the value with index $\lceil 3N/4 \rceil$. The IQ range is $Q_3 - Q_1$. The box plot shows the minimum, Q_1, median, Q_3, maximum.

Example 7.4

Determine Q_1, Q_3 and the interquartile range of the shoe sizes, then draw the box plot.

Solution:

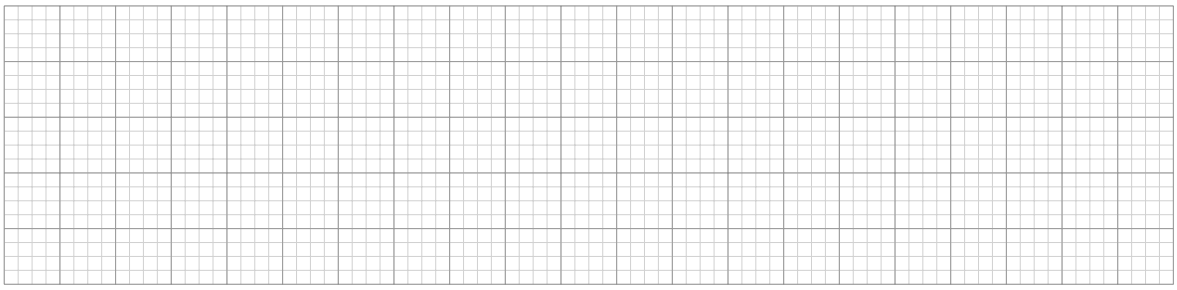




Example 7.5

Determine the quartiles of 1; 2; 2; 3; 4; 4; 4; 5. Are exactly 25% of the values less than or equal to Q_1 ?

Solution:



IV Adding or removing a value

Properties

Adding a to a series of size N gives $\overline{x}' = (N\overline{x} + a)/(N+1)$. Removing one occurrence of a, for $N > 1$, gives $\overline{x}' = (N\overline{x} - a)/(N-1)$. Use the exact sum, without intermediate rounding. Re-examine the median in the new ordered series.

Example 7.6

In the series of shoe sizes, the sum is 1217 and the median 41.

1. Add a size 36 shoe and calculate the new summary statistics.

2. Start again from the original series, remove a size 44 and repeat.

Solution:

Example 7.7

Justify that adding the mean does not change the mean. Can the range decrease when a value is added?

Solution:

V Variance and standard deviation

Definitions

$V = \dfrac{\sum n_i(x_i - \overline{x})^2}{N}$ is the variance; $\sigma = \sqrt{V}$ is the standard deviation. It measures the spread around the mean and has the same unit as the data. We can also calculate $V = \dfrac{\sum n_i x_i^2}{N} - \overline{x}^2$.

Example 7.8

Calculate the variance and standard deviation of the shoe sizes.

Solution:

Example 7.9

Compare A : 8; 10; 12 and B : 0; 10; 20 with the pairs mean-standard deviation and median-interquartile range.

Solution:

VI Linearity of the mean

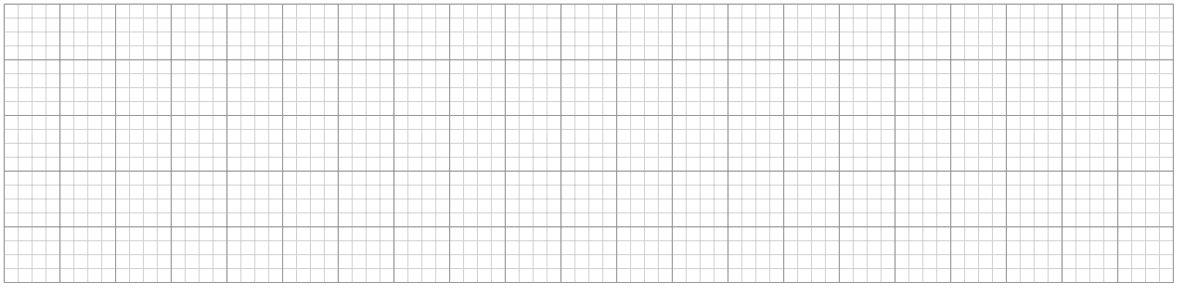
Property

Replace each x_i by $kx_i + c$ gives a mean $k\overline{x} + c$, a standard deviation $|k|\sigma$ and a variance k^2V. Adding a constant changes neither the variance nor the standard deviation.

Example 7.10

Initial mean 12 /20, standard deviation 3. Calculate the new statistics for three independent changes, without capping the marks: add 1 ,5; multiply by 0 ,8; multiply by 1 ,1 then add 0,5.

Solution:



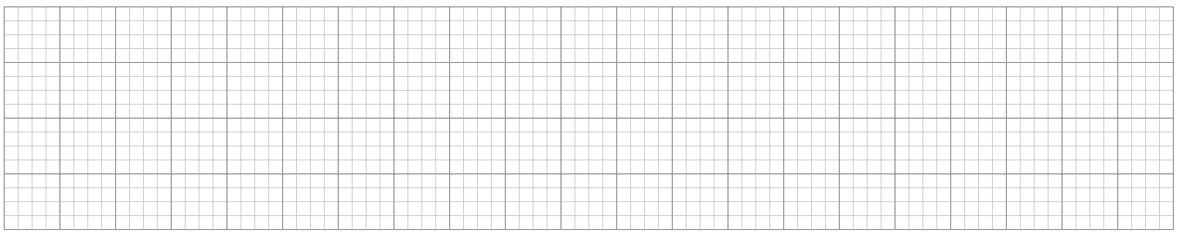
```
from math import sqrt

def moyenne_ecart_type(valeurs):
    n = len(valeurs)
    if n == 0:
        raise ValueError("liste vide")
    m = sum(valeurs)/n
    v = sum((x-m)**2 for x in valeurs)/n
    return m, sqrt(v)
```

Example 7.11

Explain the program; predict its result for `[8, 10, 12]`.

Solution:



VII Continuous data grouped into classes

Definitions and loss of information

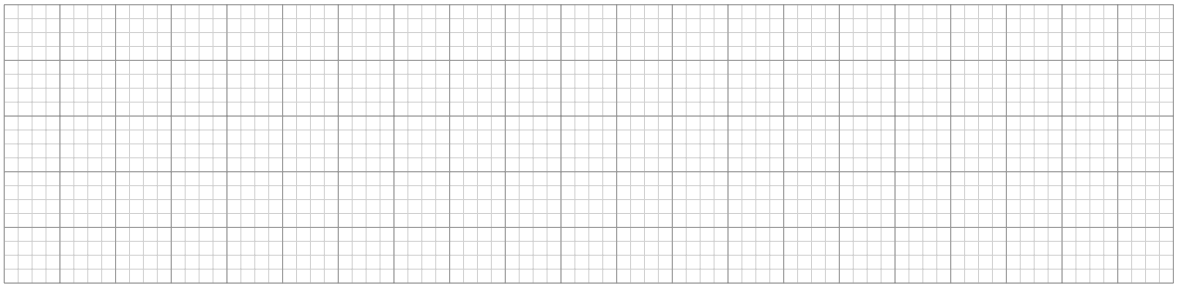
A continuous quantitative variable can take all the values of an interval. Here, the observations are grouped into classes of equal width; the midpoint of $[a; b[$ equals $(a+b)/2$. If the means m_i of the classes are known, the exact overall mean is $\sum n_i m_i / N$. Otherwise, replacing them with the class midpoints gives an estimate justified by a uniform model within each class.

Example 7.12: Heights · booklet data

Complete, estimate the mean from the class midpoints and determine the median class.

Class (cm)	$[150; 160[$	$[160; 170[$	$[170; 180[$	$[180; 190[$	$[190; 200[$
Count	8	22	35	28	7
Frequency (%)	.....	.....	.....	.....	.....
Centre (cm)	.....	.....	.....	.....	.....
ICC	.....	.....	.....	.....	.....

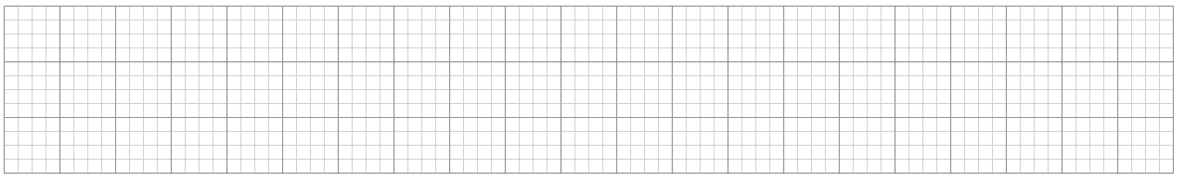
Solution:



Example 7.13

If the exact class means were 154; 166; 174; 184; 196 cm, what would the overall mean be?

Solution:



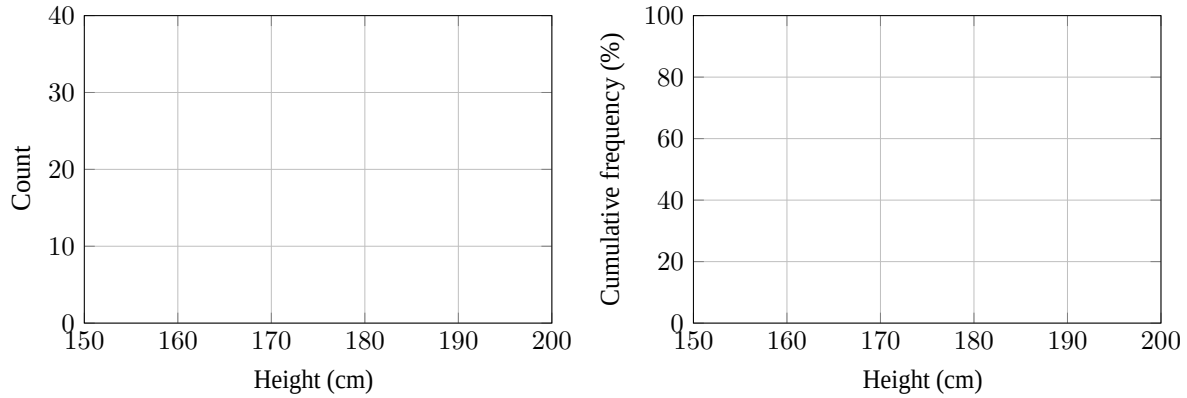
VIII Histogram and cumulative frequencies

Representations

The area of each rectangle in a histogram is proportional to the frequency. For equal class widths, the heights can be the frequencies. The cumulative frequency polygon joins the points (upper bound; cumulative frequency), starting at the first lower bound with 0.

Example 7.14

Draw the two graphs for the heights in the previous example.



Estimate the median

Assuming a uniform distribution in [170; 180[, the cumulative frequency increases linearly from 30% to 65%. Find the x-value corresponding to 50%.

Example 7.15

Calculate this estimate and state its scope.

Solution: